

Mark _____ /25

Student ID: _____

Student Name: _____

Answer All Questions:

Instructor use only

Outcomes	1	2	3	4	5	6
Covered	✓	✓	✓	NA	✓	✓

Question	1	2	3	4	5	6
Covered Outcome	6	1	1	1	6	6
Allocated mark	4	4	2	5	4	6
Earned mark						

Total	1	6	Exam
	12	13	25

Instructions: Answer all the following questions.

Question 1: Apply the Dynamic programming algorithm and give the optimal solution to the matrix chain multiplication ABCD where the order for matrix.

$A = 4 \times 5$, $B = 5 \times 8$, $C = 8 \times 3$, $D = 3 \times 7$.

Note: Provide the best order of matrix multiplication and the number of operations. [4 pts]

Answer:

	A	B	C	D
A				
B				
C				
D				

Minimum operations = _____, Best order of multiplication: _____

Question 2: Hash Table

(A): Insert integer keys $A = [4, 13, 49, 6, 40, 33]$ in order into a hash table of size 9 using the hash functions $h_1(k) = k \bmod 9$.

Collisions should be resolved via chaining, where collisions are stored at the end of a chain. Fill in the hash table after all keys have been inserted. [2 pts]

0	1	2	3	4	5	6	7	8

(B): Collisions should be resolved via double hashing, and $h_2(k) = 1 + (k \bmod 5)$. Fill in the hash table after all keys have been inserted. [2 pts]

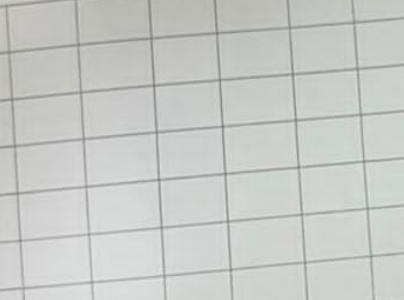
0	1	2	3	4	5	6	7	8

Question 3: Count the total primitive operations of the following algorithm [2 pts]

Algorithm $arrayMax(A, n)$ # operations
 $currentMax \leftarrow A[0]$
 $Temp \leftarrow A[0]$
 for $i \leftarrow 1$ to $n - 1$ do
 if $A[i] > currentMax$ then
 $currentMax \leftarrow A[i]$
 { increment counter i }
 return $currentMax$

The Answer: _____

(5 pts)

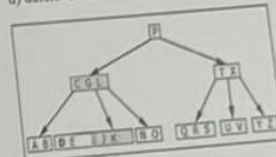
- Question 5: Using the dynamic programming algorithm, find the edit distance between two strings $X = \text{ACHAED}$ and $Y = \text{ABCADT}$.
- 
- A blank 10x10 grid is provided for the dynamic programming algorithm. The grid is tilted slightly to the right. It consists of 10 columns and 10 rows of squares, intended for calculating the edit distance between the strings X = ACHAED and Y = ABCADT.

[4 pts]

[1 pts]

- (1 gal)

[2 pts]



1

